Version 7.4.8

PVsyst - Simulation report

Grid-Connected System

Project: HW1_d

Variant: New simulation variant

Sheds on ground

System power: 7.20 kWp

Tehran - Iran

Author

PVsyst V7.4.8
VC0, Simulation date:
10/25/25 21:57
with V7.4.8

Project summary

Geographical Site
Tehran
Iran

Situation

Latitude	35.69 °N
Longitude	51.42 °E
Altitude	1178 m
Time zone	UTC+3.5

Project settings

Albedo	0.20

Weather data
Tehran
Meteonorm 8.1 (1988-2000), Sat=19% - Synthetic

System summary

Grid-Connected System

Sheds on ground

PV Field Orientation
Fixed plane
Tilt/Azimuth 36 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	16 units
Pnom total	7.20 kWp

Inverters

Nb. of units	1 unit
Pnom total	6.00 kWac
Pnom ratio	1.200

Results summary

Produced Energy	12420.28 kWh/year	Specific production	1725 kWh/kWp/year	Perf. Ratio PR	81.55 %

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General parameters

Grid-Connected System

Sheds on ground

PV Field Orientation

Orientation	
Fixed plane	
Tilt/Azimuth	36 / 0 °

Sheds configuration		
Nb. of sheds	6	units
Sizes		
Sheds spacing	2.60	m
Collector width	1.05	m
Ground Cov. Ratio (GCR)	40.2	%
Top inactive band	0.02	m
Bottom inactive band	0.02	m
Shading limit angle		
Limit profile angle	19.8	°

Models used	
Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	VSUN
Model	VSUN-450-144-MH
(Original PVsyst database)	
Unit Nom. Power	450 Wp
Number of PV modules	16 units
Nominal (STC)	7.20 kWp
Modules	2 string x 8 In series
At operating cond. (50°C)	
Pmpp	6.53 kWp
U mpp	298 V
I mpp	22 A

Inverter

Manufacturer	Huawei Technologies
Model	SUN2000-6KTL-L1
(Original PVsyst database)	
Unit Nom. Power	6.00 kWac
Number of inverters	1 unit
Total power	6.0 kWac
Operating voltage	90-560 V
Pnom ratio (DC:AC)	1.20
Power sharing within this inverter	

Total PV power

Nominal (STC)	7 kWp
Total	16 modules
Module area	35.3 m²

Total inverter power

Total power	6 kWac
Number of inverters	1 unit
Pnom ratio	1.20

Array losses

Thermal Loss factor

Module temperature according to irradiance		
Uc (const)	20.0	W/m²K
Uv (wind)	0.0	W/m²K/m/s

DC wiring losses

Global array res.	227	mΩ
Loss Fraction	1.5	% at STC

Module Quality Loss

Loss Fraction	-0.8 %

Module mismatch losses

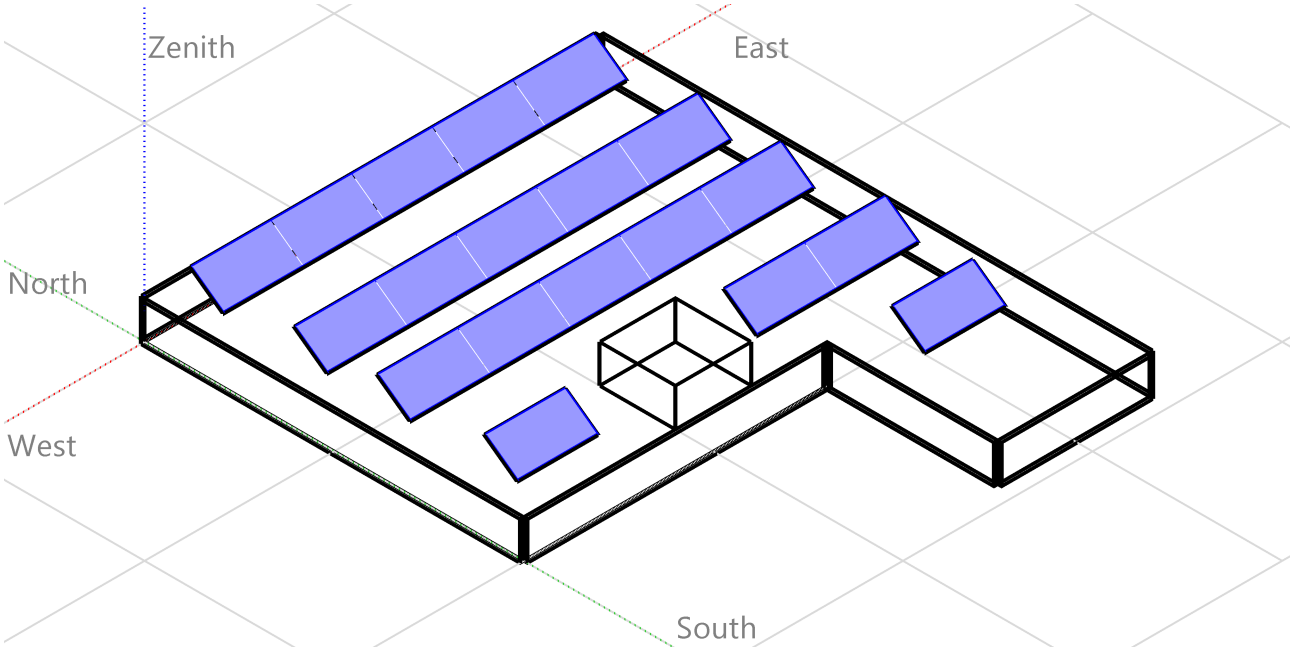
Loss Fraction	2.0 % at MPP

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.403	0.000

Near shadings parameter

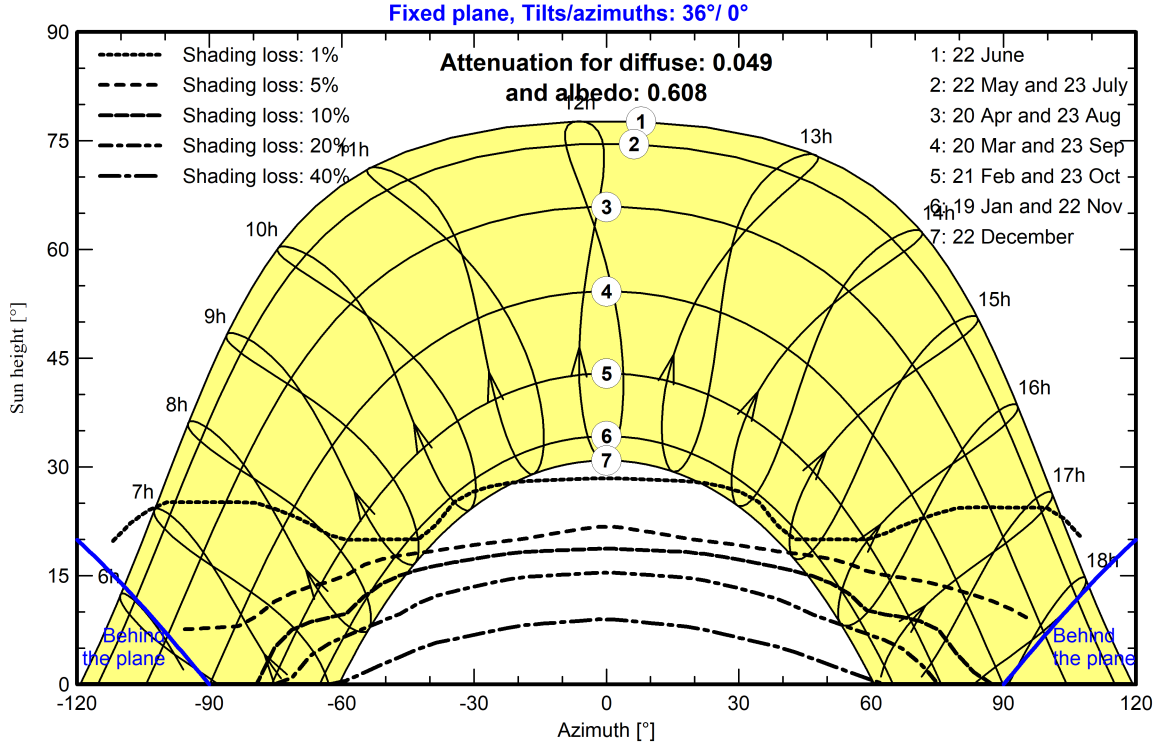
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 36°/ 0°

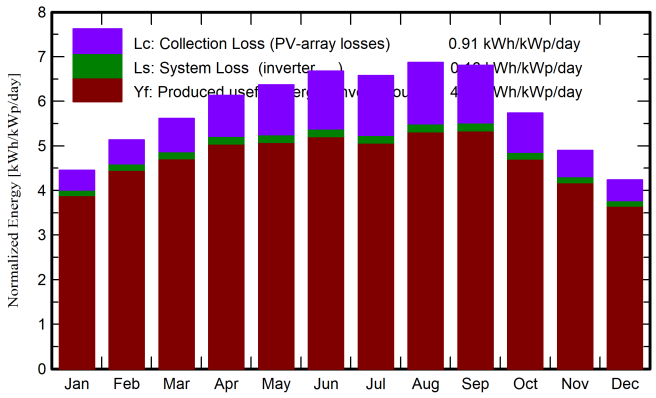


Main results

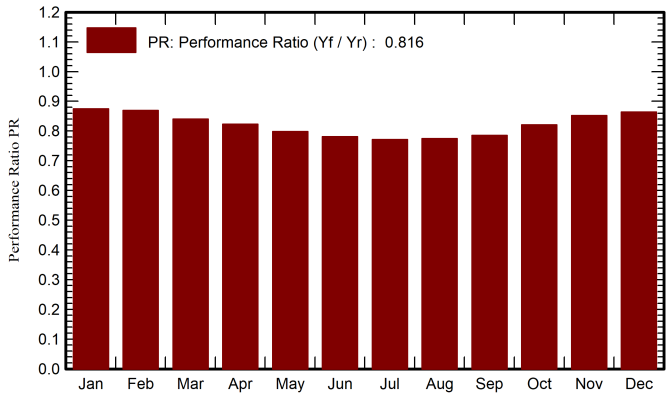
System Production

Produced Energy	12420.28 kWh/year	Specific production	1725 kWh/kWp/year
		Perf. Ratio PR	81.55 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

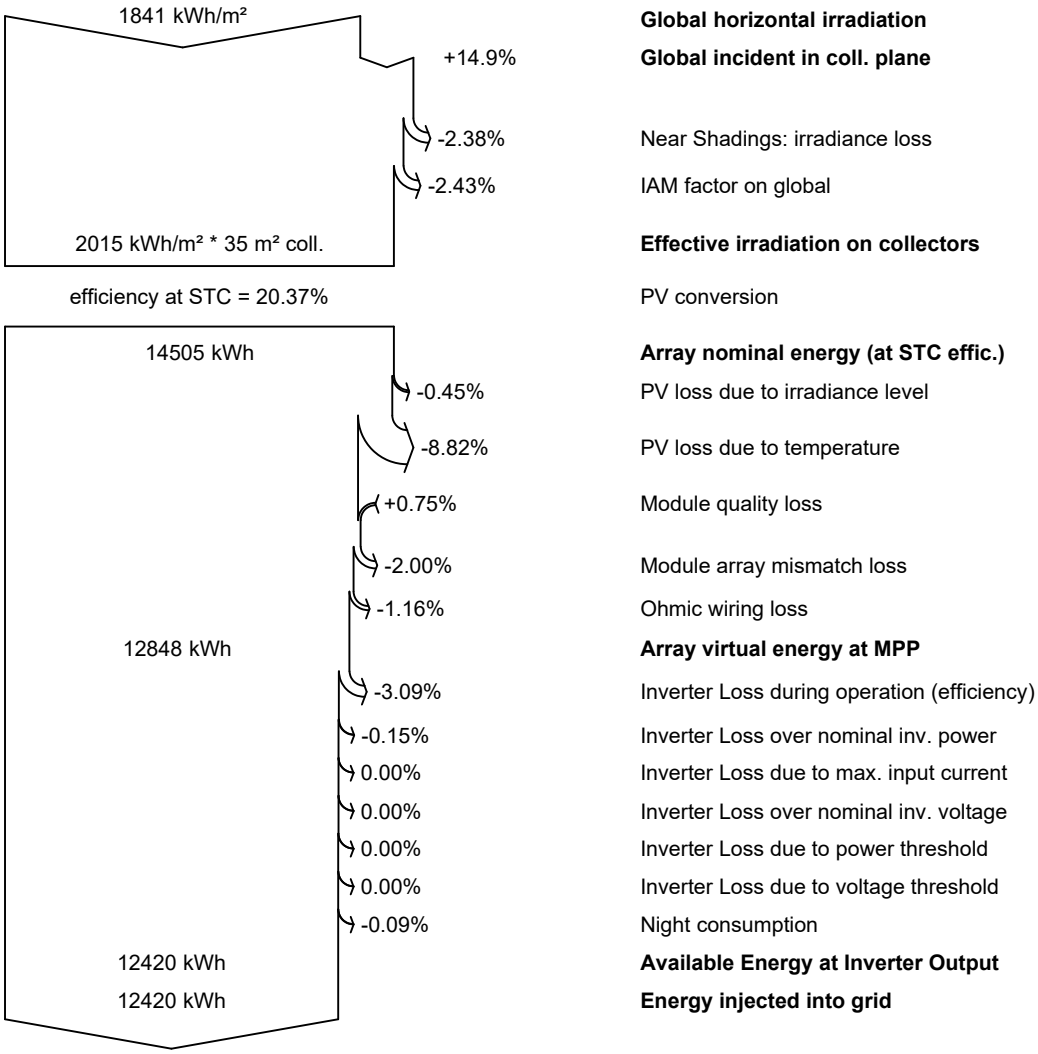
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	83.3	29.83	5.08	138.1	131.2	897	869	0.874
February	99.6	34.42	7.26	143.7	138.1	928	899	0.869
March	142.5	54.73	12.75	174.2	166.5	1089	1054	0.840
April	174.2	65.79	17.59	184.1	175.0	1127	1091	0.823
May	209.5	70.09	23.73	197.5	187.1	1174	1135	0.798
June	224.7	73.09	28.75	200.3	189.3	1163	1126	0.781
July	222.3	77.43	31.66	203.9	193.0	1170	1132	0.771
August	209.7	65.80	30.66	213.0	202.8	1228	1188	0.775
September	174.8	45.45	26.24	204.2	195.3	1193	1154	0.785
October	131.8	43.74	19.97	178.0	171.3	1086	1051	0.820
November	91.8	31.48	11.30	147.0	140.2	932	902	0.852
December	76.5	26.05	6.44	131.4	124.9	844	817	0.863
Year	1840.6	617.91	18.52	2115.2	2014.7	12830	12420	0.816

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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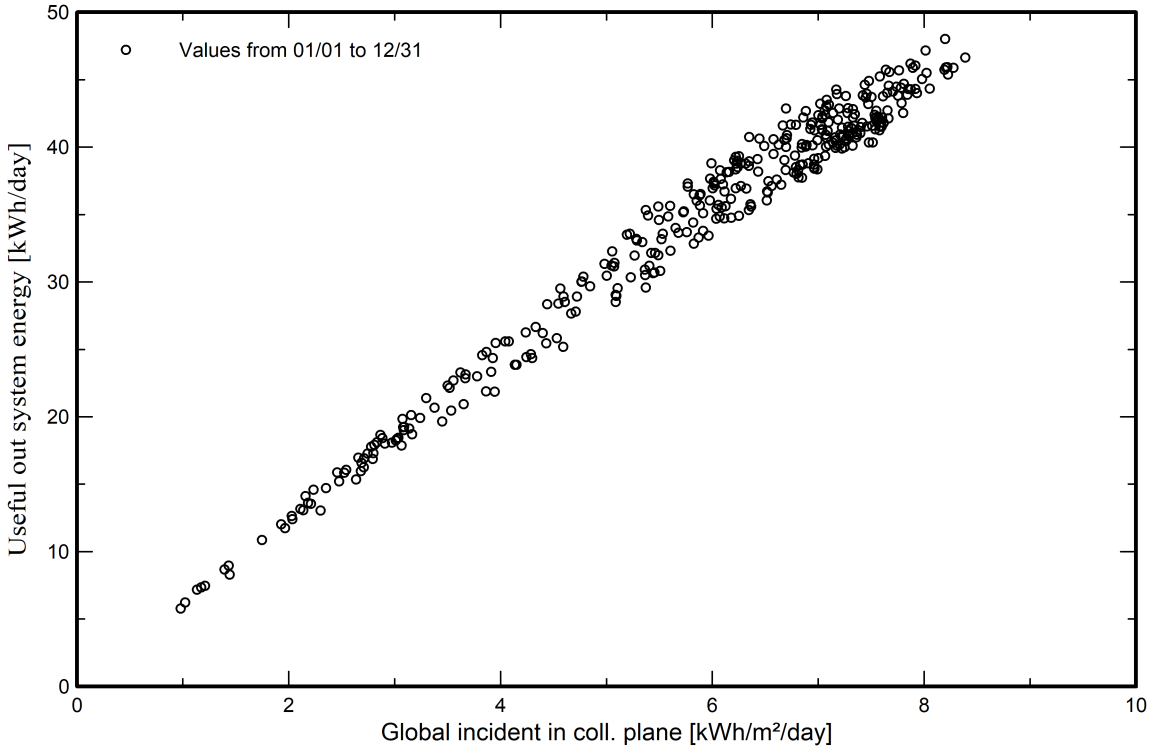
Loss diagram



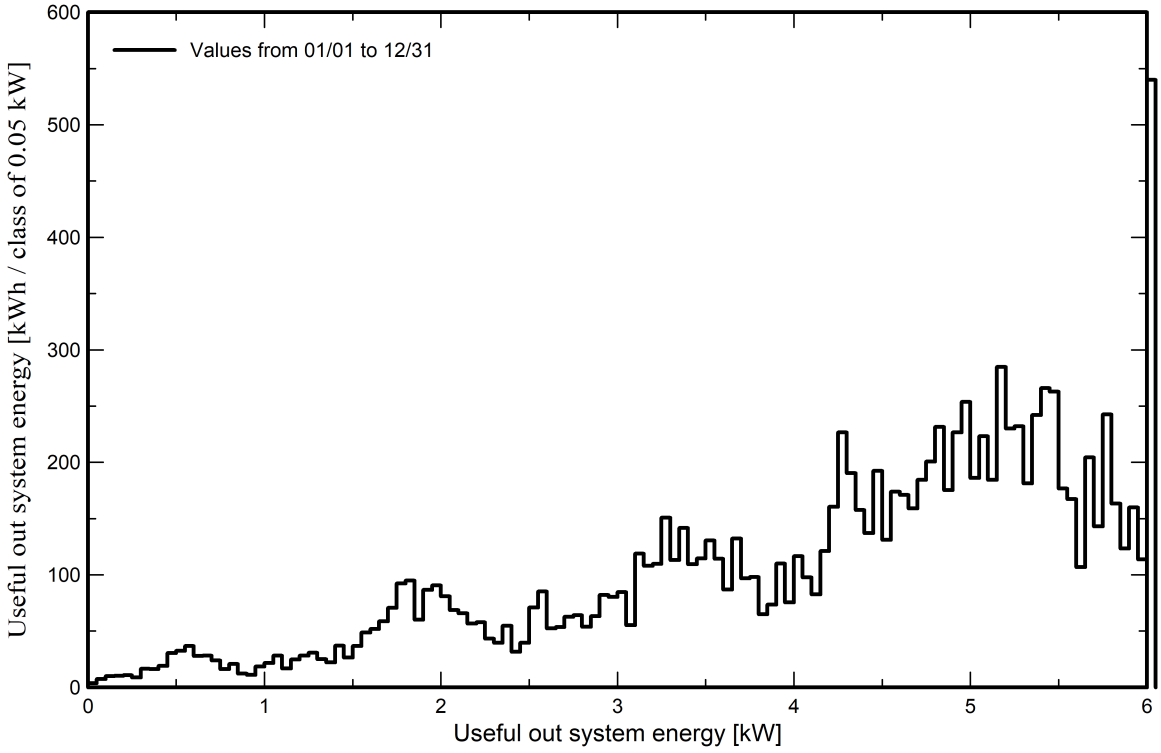
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Predef. graphs

Daily Input/Output diagram

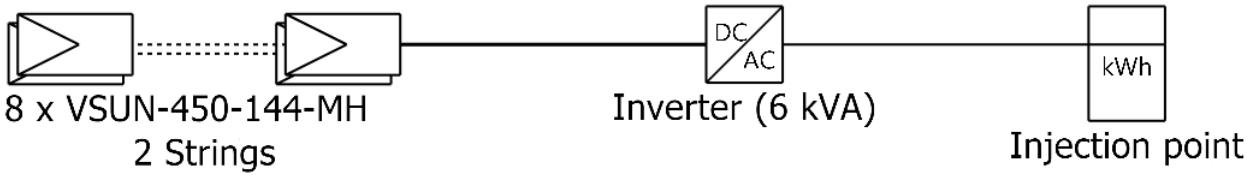


System Output Power Distribution



Single-line diagram

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PV module	VSUN-450-144-MH
Inverter	SUN2000-6KTL-L1
String	8 x VSUN-450-144-MH

	HW1_d	
	VC0 : New simulation variant	10/25/25